

**MSTAR PRACTICE WORKSHEET****Lesson 8-1: Understand Equations and Their Solutions**

Grade 6 Mathematics · Standard 6.AT.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

**Part A — Skills Check**

One point each.

**Question 1** SELECTED RESPONSE — CORRECT: A

Which equation represents: 'A number plus 12 equals 30'?

- ☒ A  $n + 12 = 30$
- ☐ B  $n - 12 = 30$
- ☐ C  $12n = 30$
- ☐ D  $n / 12 = 30$

'Plus' means addition, so the equation is  $n + 12 = 30$ .

**Question 2** SELECTED RESPONSE — CORRECT: A

Which equation represents: 'Seven less than a number is 18'?

- ☒ A  $n - 7 = 18$
- ☐ B  $7 - n = 18$
- ☐ C  $n + 7 = 18$
- ☐ D  $n / 7 = 18$

'Seven less than a number' means start with the number and subtract 7:  $n - 7 = 18$ .

**Question 3**    SELECTED RESPONSE — CORRECT: A

Which equation represents: 'Three times a number equals 21'?

- ☒ A  $3n = 21$
- ☐ B  $n + 3 = 21$
- ☐ C  $n - 3 = 21$
- ☐ D  $n / 3 = 21$

'Three times' means multiply by 3, so the equation is  $3n = 21$ .

**Question 4**    SELECTED RESPONSE — CORRECT: B

Is  $n = 3$  a solution of  $n + 5 = 9$ ?

- ☐ A Yes, because 3 is smaller than 9
- ☒ B No, because  $3 + 5 = 8$ , which does not equal 9
- ☐ C Yes, because  $3 + 5$  is close to 9
- ☐ D You cannot tell without solving first

Substitute the value and check whether the statement is true:  $3 + 5 = 8$ , and  $8 \neq 9$ , so 3 is not a solution.

**Question 5**    SELECTED RESPONSE — CORRECT: A

Which equation represents: 'A number divided by 4 equals 8'?

- ☒ A  $n / 4 = 8$
- ☐ B  $4n = 8$
- ☐ C  $n - 4 = 8$
- ☐ D  $n + 4 = 8$

'Divided by 4' means division:  $n / 4 = 8$ .

**Question 6**

SELECTED RESPONSE — CORRECT: A

A detective found some fingerprints on Monday and 14 more on Tuesday, for a total of 30. Which equation represents this?

**A**  $f + 14 = 30$

**B**  $f - 14 = 30$

**C**  $14f = 30$

**D**  $f / 14 = 30$

The detective's Monday fingerprints (f) plus 14 more equals 30:  $f + 14 = 30$ .

**Part B — Test-Format Questions**

Scoring notes and distractor rationales below each item.

**Question 7** TWO-PART QUESTION**Part A — correct answer: B**

Marisol tests values in the equation  $7 + m = 23$ . Which value of  $m$  makes the equation true?

- ☐ A  $m = 9$
- ☒ B  $m = 16$
- ☐ C  $m = 23$
- ☐ D  $m = 30$

Substitute each value for the variable. When  $m = 16$ , the left expression is  $7 + 16 = 23$  and the right expression is 23, so both sides balance. That makes 16 the solution.

- **A:** Substituting 9 gives 7 plus 9, which reaches only 16.
- **C:** That is the total, not the missing addend. Seven plus 23 comes to 30.
- **D:** That adds 7 and 23. To undo an addition of 7, subtract it from 23.

**Part B — correct answer: C**

Which reasoning best justifies that answer?

- ☐ A Adding the two numbers written in the equation,  $7 + 23$ , gives the value of  $m$ .
- ☐ B The equal sign means the answer is always the number written to the right of it.
- ☒ C Substituting 16 for  $m$  gives  $7 + 16 = 23$ , so both sides of the equation have the same value.
- ☐ D Any number can replace  $m$  because  $m$  is a variable, and variables can change.

A value is a solution only when substituting it makes the left expression equal the right expression. Substituting 16 gives  $23 = 23$ , so the equation is true. Adding 7 and 23 gives 30, and  $7 + 30 = 37$ , which does not balance.

**Question 8** SELECT ALL THAT APPLY — CORRECT: A, B, C

Select ALL statements that are true about equations and their solutions.

- ☒ A An equation is a math sentence that uses an equal sign to show two expressions have the same value.
- ☒ B You can test a value by substituting it for the variable and checking whether both sides are equal.
- ☒ C The value  $n = 6$  is a solution of  $n + 9 = 15$  because  $6 + 9 = 15$ .
- ☐ D The expression  $4y + 2$  is an equation because it contains a variable.
- ☐ E The value  $x = 15$  is a solution of  $x - 4 = 9$  because  $15 - 4 = 9$ .

The first three statements are true. The fourth is false because  $4y + 2$  has no equal sign, so it is an expression, not an equation. The fifth is false because  $15 - 4 = 11$ , not 9; the value that makes  $x - 4 = 9$  true is  $x = 13$ , since  $13 - 4 = 9$ .

## Part C — Show Your Thinking

Model answer and scoring guidance.

### Question 9 WRITTEN RESPONSE · 2 POINTS

Write a real-world situation that can be modeled by the equation  $n / 5 = 9$ . Explain what  $n$  represents and verify your equation makes sense.

**Model answer:** The equation  $n / 5 = 9$  describes one amount being shared into 5 equal groups with 9 in each group, so a matching story could be  $n$  stickers split evenly among 5 friends with each friend getting 9. Since 5 groups of 9 rebuild the whole,  $n = 9 \times 5 = 45$ , and  $45 \div 5 = 9$  checks the story.

*Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.*

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Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.